BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The male reproductive system

The written notes for the male reproductive lecture: the scrotum and testis, the sperm duct pathway, the sperm cell, the accessory glands and penis, and the disorders of each structure. Module 5 competencies 141 to 145, with eight worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Before the competencies: the four groups of parts

In your course notes: Module 5 packet, Chapter 2, competency 141 (page 14).

Dr. Rennie's video for this part: Male reproductive anatomy concept video, chapter Male reproductive system overview, and Hormone control at puberty.

The male reproductive system makes sperm and delivers them. Its parts sort into four groups: the gonads, a duct system, the accessory glands, and the supporting structures. Learn the groups first, because every structure in this chapter belongs to one of them, and the order of the duct system is the order sperm travel.

Group	What it includes
Testes	The male gonads. They produce sperm and secrete hormones (the interstitial cells make testosterone)
Duct system	The epididymis, ductus deferens, ejaculatory ducts, and urethra, which store, mature, and carry sperm
Accessory glands	The seminal vesicles, prostate, and bulbourethral glands, which add fluid to form semen
Supporting structures	The scrotum and the penis

- Competency 141 covers the testes and the scrotum. Competency 142 covers the duct system. Competency 143 covers the sperm cell itself. Competency 144 covers the accessory glands and the penis. Competency 145 puts the disorders on the structures.
- Semen is sperm plus the secretions of the three accessory glands. The sperm are only a small part of it.

Competency 141. Scrotum and testis structure

In your course notes: Module 5 packet, Chapter 2, competency 141 (page 14).

Dr. Rennie's video for this part: Male reproductive anatomy concept video, chapter Scrotum temperature control muscles, and Testis structure and sperm production.

You should be able to: Identify the scrotum, dartos and cremaster muscles, tunica layers, seminiferous tubules,

sustentacular and interstitial cells, rete testis, and efferent ductules.

This section covers the gonads, the testes, and the pouch that holds them. The testes sit outside the body because sperm production needs a temperature a few degrees below the body core, and two muscles in the scrotum adjust that temperature by moving the testes closer to or farther from the body.

The scrotum and its two muscles

The scrotum is a pouch of skin and subcutaneous tissue that hangs behind the penis, outside the pelvic cavity. A ridge down its middle, the raphe, marks where the scrotal septum divides the inside into two compartments, one for each testis.

Muscle	Tissue type	Where it is	What it does when it contracts
Dartos muscle	Smooth muscle	A thin layer in the subcutaneous tissue of the scrotum, under the skin	Wrinkles and tightens the scrotal skin, which cuts heat loss
Cremaster muscle	Skeletal muscle	Comes off the internal oblique muscle of the abdominal wall, travels down in the spermatic cord, and loops around each testis	Pulls the testis up toward the warmer body wall

- The two muscles are different tissues. Dartos is smooth muscle in the scrotal wall. Cremaster is skeletal muscle in the spermatic cord.
- In warm conditions both muscles relax and the testes hang lower, away from the body.
- The two structures that set the temperature of the testes are the two muscles, the dartos and the cremaster, not the tunics.

The coverings of the testis: the two tunics

Tunica vaginalis

The outer tunic, a serous membrane with a parietal layer and a visceral layer. It comes from a pouch of peritoneum that the testis brought with it when it moved down into the scrotum before birth, and it covers the front and sides of the testis (it partly covers the testis, not the whole thing). It lets the testis move smoothly inside the scrotum.

Tunica albuginea

The inner tunic, a tough white capsule of dense connective tissue that sits directly on the testis. It sends thin walls (septa) inward that divide the testis into a few hundred wedge-shaped lobules.

Inside the testis

Each lobule holds tightly coiled seminiferous tubules. These tubules are the only place sperm are made. On a slide, a seminiferous tubule is a round or oval profile with a thick wall of cells around a small central lumen. The developing sperm cells are stacked in that wall, the least mature at the outer edge and the most mature next to the lumen.

Sustentacular cells (Sertoli cells)

Tall cells inside the wall of each seminiferous tubule, reaching from the outer edge of the tubule to its lumen. They surround, support, and nourish the developing sperm cells. Tight junctions between neighboring sustentacular cells form the blood-testis barrier, which keeps substances in the blood away from the developing sperm.

Interstitial cells (Leydig cells)

Cells in the connective tissue between the tubules, outside the tubule walls. They produce testosterone.

Straight tubules

Short tubules that carry sperm from the seminiferous tubules toward the back of the testis.

Rete testis

A network of small channels in the connective tissue at the posterior side of the testis. It collects the sperm from the tubules.

Efferent ductules

A set of tiny ducts that leave the rete testis, carry the sperm out of the testis, and deliver them into the head of the epididymis.

- Memory tip: the names tell you where to look on the slide. Sustentacular cells sustain the sperm, so they sit inside the tubule wall with the sperm. Interstitial cells sit in the interstitial spaces, between the tubules.
- Interstitial cells sit between the tubules, not in the tubule wall. The sustentacular cells are the ones in the wall.

- Inside the testis the sperm go seminiferous tubules, straight tubules, rete testis, efferent ductules, epididymis. The next competency picks up from there.

The scrotum and the testis, from the outside in

Structure	Where it is	What it does
Scrotum	Skin pouch behind the penis, outside the pelvic cavity	Holds the testes and keeps them cooler than the body core; the scrotal septum gives each testis its own compartment
Dartos muscle	Smooth muscle in the subcutaneous tissue of the scrotum	Wrinkles the scrotal skin to reduce heat loss
Cremaster muscle	Skeletal muscle in the spermatic cord, looping around the testis	Raises the testis toward the body
Tunica vaginalis	Serous membrane over the front and sides of the testis	Lets the testis move smoothly inside the scrotum
Tunica albuginea	Dense white capsule directly on the testis	Encloses the testis and divides it into lobules
Seminiferous tubules	Coiled tubules inside each lobule	Produce sperm
Sustentacular cells	Inside the wall of each seminiferous tubule	Support and nourish developing sperm; form the blood-testis barrier
Interstitial cells	In the connective tissue between the tubules	Produce testosterone
Rete testis	Network of channels at the posterior side of the testis	Collects sperm from the tubules
Efferent ductules	Small ducts between the rete testis and the epididymis	Carry sperm out of the testis into the epididymis

Competency 142. Sperm duct system and pathway

In your course notes: Module 5 packet, Chapter 2, competency 142 (page 16).

Dr. Rennie's video for this part: Male reproductive anatomy concept video, chapter Ducts: sperm to ejaculation.

You should be able to: Trace the path of sperm from the seminiferous tubules to the outside, naming the epididymis, ductus deferens, ejaculatory duct, urethra, and spermatic cord contents.

Only the seminiferous tubules make sperm. Every duct after them stores, matures, or carries the sperm. The route runs in one direction: up out of the scrotum, into the pelvis behind the bladder, and then down through the prostate and the penis to the outside.

The path of sperm, from the testis to the outside

1. Seminiferous tubules: sperm are made in the walls of the tubules inside the testis.
2. Rete testis: straight tubules carry the sperm into this network at the back of the testis.
3. Efferent ductules: small ducts carry the sperm out of the testis.
4. Epididymis: sperm enter the head, are stored, and mature as they move to the tail.
5. Ductus deferens: sperm travel up the spermatic cord, through the inguinal canal, over the ureter, and behind the bladder to the ampulla.
6. Ejaculatory duct: the ductus deferens joins the duct of the seminal vesicle and passes through the prostate.

7. Prostatic urethra: sperm enter the urethra inside the prostate.
8. Membranous urethra: sperm pass through the pelvic floor.
9. Spongy urethra: sperm pass through the corpus spongiosum of the penis.
10. External urethral orifice: semen leaves the body at the tip of the glans.

- Memory tip: SEVEN UP gives the main stops in order: Seminiferous tubules, Epididymis, Vas deferens, Ejaculatory duct, (N is for nothing), Urethra, Penis.
- Only the seminiferous tubules make sperm. The epididymis stores and matures them.

Epididymis

The epididymis is a comma-shaped structure on the posterior side of each testis. Its head sits at the top, where the efferent ductules enter. Its body runs down the back of the testis, and its tail sits at the bottom. Inside is a single duct about 6 meters long, coiled tightly enough to fit in a few centimeters. Sperm are stored here, and they gain the ability to swim as they move through it.

Ductus deferens

The ductus deferens (vas deferens) begins at the tail of the epididymis. It is a long duct with a thick wall of smooth muscle, so it feels like a firm cord through the scrotal skin. It climbs in the spermatic cord, passes through the inguinal canal into the pelvic cavity, crosses over the ureter, and runs down the posterior surface of the bladder. Its last part widens into the ampulla of the ductus deferens.

Ejaculatory duct

Behind the bladder, each ampulla joins the duct of a seminal vesicle. The two together form the short ejaculatory duct. There is one on each side. Each ejaculatory duct passes through the prostate and empties into the prostatic urethra. It does not end in the bladder.

The three parts of the male urethra

The male urethra carries both urine and semen. It has three parts, named for what they pass through.

Part	Where it runs	Notes
Prostatic urethra	Through the prostate, just below the bladder	First part; the ejaculatory ducts and the prostate's ducts empty into it
Membranous urethra	Through the muscles of the pelvic floor (the urogenital diaphragm), in the perineum	Short middle part
Spongy urethra (penile urethra)	Through the corpus spongiosum of the penis	Longest part; opens at the external urethral orifice on the glans

Spermatic cord

The spermatic cord is the bundle that suspends each testis and carries its duct, vessels, and nerves between the scrotum and the abdomen. It passes through the inguinal canal, a slanted passage through the lower anterior abdominal wall. The cord holds these structures:

- The ductus deferens.
- The testicular artery, a branch of the abdominal aorta.
- The pampiniform plexus, a network of veins wrapped around the testicular artery. It cools the blood flowing in to the testis, and its veins join to form the testicular vein.
- Lymphatic vessels and nerves.
- The cremaster muscle and layers of fascia that wrap the whole cord.

If you cut the cord across, you find the ductus deferens, blood vessels, nerves, and the cremaster muscle inside it.

The ducts, in order

Structure	Where it is	What it does
Epididymis	Posterior side of the testis	Stores sperm; sperm gain the ability to swim
Ductus deferens	From the epididymis, up the spermatic cord, through the inguinal canal, behind the bladder	Carries sperm into the pelvis
Ejaculatory duct	Inside the prostate	Joins the ductus deferens and seminal vesicle duct; empties into the prostatic urethra
Prostatic urethra	Inside the prostate	First part of the urethra
Membranous urethra	Through the pelvic floor muscles	Short middle part of the urethra
Spongy urethra	Inside the corpus spongiosum of the penis	Longest part; opens at the external urethral orifice
Spermatic cord	From the scrotum through the inguinal canal	Carries the ductus deferens, vessels, and nerves of the testis

Competency 143. Sperm cell structure

In your course notes: Module 5 packet, Chapter 2, competency 143 (page 17).

Dr. Rennie's video for this part: Male reproductive anatomy concept video, chapter Cryptorchidism and sperm basics.

You should be able to: Identify the head, acrosome, neck, middle piece, principal piece, and end piece of a sperm cell.

A mature sperm cell (spermatozoon) is built for one job: reaching an oocyte and delivering its chromosomes. It has two main parts, a head and a tail. The tail is a long flagellum. From front to back, the tail has four parts: the neck, the middle piece, the principal piece, and the end piece. These notes count the neck as part of the tail, so the tail starts right behind the head.

The head and its cap

Head

A flattened oval at the front of the cell. Most of it is the nucleus, which holds 23 chromosomes, half the number in a body cell. There is almost no cytoplasm.

Acrosome

A cap that covers the front of the nucleus. It is a sac of enzymes that lets the sperm digest a path through the layers around the oocyte.

The four parts of the tail

Neck

The short, narrow part just behind the head. It holds the centrioles, and the core of microtubules that runs the length of the tail starts here.

Middle piece

The first long stretch of the tail, packed with mitochondria wound in a spiral around the core of the tail. They make the ATP that powers the tail. This is the only part packed with mitochondria.

Principal piece

The longest part of the tail. It does most of the whipping that moves the cell forward.

End piece

The short, thin tip at the very end of the tail, tapering to a point.

The parts of a sperm cell, front to back

Part	Where it is	What it holds or does
Head	Front of the cell	Holds the nucleus with 23 chromosomes
Acrosome	Cap over the front of the head	Holds enzymes that help the sperm get through the layers around the oocyte
Neck	Just behind the head	Holds the centrioles; the tail core starts here
Middle piece	First long stretch of the tail	Packed with mitochondria that power movement
Principal piece	Most of the tail	Longest part; drives the swimming
End piece	Tip of the tail	Short, thin final part

- Memory tip: name the parts front to back: head with its acrosome cap, then neck, middle piece, principal piece, end piece. The middle piece is the only part packed with mitochondria.
- The mitochondria go in the middle piece, not in the head or the end piece.
- On a slide of the testis, you see sperm heads and tails crowded in the lumen of the seminiferous tubules. On a sperm smear, the head is the dark, densely stained oval, and the tail is the long thin line behind it.

Competency 144. Accessory glands and penis

In your course notes: Module 5 packet, Chapter 2, competency 144 (page 18).

Dr. Rennie's video for this part: Male reproductive anatomy concept video, chapter Semen glands, urethra, and penis.

You should be able to: Name the seminal vesicles, prostate, and bulbourethral glands and identify the root, body, glans, prepuce, and three erectile columns of the penis.

The accessory glands

Sperm are only a small part of semen. Three sets of accessory glands add most of the fluid. The mix of sperm and these secretions is semen. Where each gland sits tells you where its fluid joins the duct system.

Seminal vesicles

A pair of glands on the posterior surface of the bladder, beside the ampullae of the ductus deferens. The duct of each seminal vesicle joins the ductus deferens to form the ejaculatory duct. Their fluid is alkaline and rich in fructose, a sugar the sperm use for energy. It makes up about 60 percent of the volume of semen, the largest share of any gland.

Prostate

A single gland about the size of a walnut. It sits just below the bladder and wraps around the prostatic urethra. Its back surface lies against the rectum. Many small ducts carry its milky fluid into the prostatic urethra, where it helps the sperm stay active.

Bulbourethral glands (Cowper glands)

A pair of pea-sized glands just below the prostate, beside the membranous urethra in the pelvic floor. Their ducts open into the spongy urethra. They release a clear alkaline mucus before ejaculation that neutralizes traces of acidic urine in the urethra and lubricates the tip of the penis.

Gland	Number	Where it is	Empties into	What it adds
Seminal vesicles	Paired	Posterior surface of the bladder	Ejaculatory duct	Alkaline, fructose-rich fluid that nourishes the sperm; about 60 percent of semen
Prostate	Single	Just below the bladder, around the prostatic urethra	Prostatic urethra	Milky fluid that helps sperm stay active
Bulbourethral glands	Paired	Below the prostate, beside the membranous urethra	Spongy urethra	Clear alkaline mucus that neutralizes urine traces and lubricates

- The glands empty in the same order the sperm travel: seminal vesicles into the ejaculatory duct, prostate into the prostatic urethra, bulbourethral glands into the spongy urethra.

The three regions of the penis

The penis carries the spongy urethra and delivers semen. It has three regions, plus the fold of skin at the tip.

Root

The attached part, fixed to the pelvis in the perineum. It is made of the bulb of the penis, which is the back end of the corpus spongiosum, and the two crura, which are the back ends of the corpora cavernosa and attach to the ischial and pubic rami.

Body (shaft)

The free, movable part.

Glans penis

The enlarged tip, formed by the expanded end of the corpus spongiosum. The external urethral orifice opens on the glans.

Prepuce (foreskin)

The loose fold of skin that covers the glans. It is the part removed in circumcision.

The three erectile columns

Three columns of erectile tissue run the length of the body. Erectile tissue is a sponge of blood spaces inside connective tissue and smooth muscle. When these spaces fill with blood, the penis becomes erect.

Column	Number	Side	What it forms
Corpora cavernosa	Two, side by side	Dorsal	Their back ends form the crura of the root
Corpus spongiosum	One	Ventral	Surrounds the spongy urethra; its back end forms the bulb and its front end forms the glans

- Memory tip: in a cross section of the penis, find the urethra first. The column wrapped around it is the corpus spongiosum. The two larger columns above it are the corpora cavernosa.
- The urethra runs through the single corpus spongiosum on the ventral side, not between the two corpora cavernosa.
- The glans is an extension of the corpus spongiosum, not of the corpora cavernosa.

The regions of the penis and its erectile columns

Structure	What it is
Root	The attached part; the bulb of the penis and the two crura
Body	The free, movable shaft
Glans penis	The enlarged tip; the expanded end of the corpus spongiosum
Prepuce	The foreskin; the loose fold of skin over the glans
Corpora cavernosa	The two dorsal columns of erectile tissue; their back ends form the crura
Corpus spongiosum	The single ventral column; surrounds the spongy urethra and forms the bulb and the glans

Competency 145. Clinical anatomy of male disorders

In your course notes: Module 5 packet, Chapter 2, competency 145 (page 19).

Dr. Rennie's video for this part: Male reproductive anatomy concept video, chapter Cryptorchidism and sperm basics.

You should be able to: Relate male reproductive disorders such as cryptorchidism and benign prostatic hyperplasia to the affected structure and its position.

Most male reproductive disorders make sense once you know where the structure sits. The testes form inside the abdomen and move down into the scrotum. The prostate wraps around the urethra. The veins of the testis run up a long cord. Each of these positions explains a disorder.

The descent of the testes and cryptorchidism

Before birth, the testes develop high on the posterior abdominal wall, near the kidneys. That is why each testicular artery branches from the abdominal aorta. Late in fetal development, each testis moves down and passes through the inguinal canal into the scrotum, pulling its duct and vessels behind it. In cryptorchidism, one or both testes stop along this route, in the abdomen or in the inguinal canal. An undescended testis stays at core body temperature, which impairs sperm production and raises the risk of testicular cancer.

The prostate: benign prostatic hyperplasia and prostate cancer

The prostate sits just below the bladder and surrounds the prostatic urethra. In benign prostatic hyperplasia (BPH), the prostate enlarges without cancer, most often in the tissue right around the urethra. The enlarged gland squeezes the prostatic urethra, so the urine stream weakens and the bladder may not empty fully. BPH is common in older men. Prostate cancer most often starts in the outer, posterior part of the gland. Because the back of the prostate lies against the rectum, a clinician can feel it through the rectal wall.

The inguinal canal, the tunica vaginalis, and the pampiniform plexus

The inguinal canal is a weak spot in the abdominal wall, left by the descent of the testis. In an inguinal hernia, a loop of intestine pushes into the canal and can follow the spermatic cord down into the scrotum. In a hydrocele, serous fluid collects between the two layers of the tunica vaginalis. In a varicocele, the veins of the pampiniform plexus swell and twist. Varicoceles are more common on the left, where the testicular vein drains into the left renal vein.

Male disorders by structure and position

Disorder	Structure affected	Where it is	What changes
Cryptorchidism	Testis	Abdomen or inguinal canal, instead of the scrotum	The testis fails to descend and stays at core body temperature
Benign prostatic hyperplasia	Prostate	Below the bladder, around the prostatic urethra	Non-cancerous enlargement that squeezes the urethra
Prostate cancer	Prostate	Usually the posterior part, against the rectum	Malignant tumor of the gland
Testicular cancer	Testis	In the scrotum	Malignant tumor, usually from the sperm-forming cells of the seminiferous tubules
Hydrocele	Tunica vaginalis	Around the testis	Serous fluid collects between its two layers and the scrotum swells
Varicocele	Pampiniform plexus	Spermatic cord, above the testis	The veins swell and twist, more often on the left
Inguinal hernia	Inguinal canal	Lower anterior abdominal wall	A loop of intestine pushes into the canal along the spermatic cord

What this means for patients

- Sperm production needs a temperature a few degrees below the body core. That is why the testes sit in the scrotum and why the cremaster and dartos muscles adjust their position. In cryptorchidism, an undescended testis stays at body temperature, and if it is not corrected the warmth can damage the sperm-forming cells.
- The duct system shares its final stretch with the urinary tract. Because the urethra passes straight through the prostate, an enlarged prostate squeezes the urethra and makes urination difficult. This is how a reproductive organ produces a urinary symptom. BPH squeezes the urethra, not the bladder.
- Hydrocele is fluid around the testis, between the layers of the tunica vaginalis. Varicocele is swollen veins in the spermatic cord above the testis. Do not swap them.
- The back of the prostate lies against the rectum, which is why a clinician can feel the posterior part of the gland, where prostate cancer usually starts, through the rectal wall.
- The inguinal canal is the passage the testis took on its way down, which is why an inguinal hernia can follow the spermatic cord into the scrotum.

Worked questions

Eight questions in the style of the lecture exam, at least one for each competency. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 141

Which cells of the testis produce testosterone?

A. Sustentacular cells

B. Cells lining the rete testis

C. Interstitial cells

D. Cells of the tunica albuginea

Answer: C. Interstitial cells (Leydig cells) sit in the connective tissue between the seminiferous tubules, outside the tubule walls, and they make testosterone.

Why the others are wrong

- **A** Sustentacular cells (Sertoli cells) sit inside the tubule wall. They support and nourish the developing sperm and form the blood-testis barrier; they do not make testosterone.
- **B** The rete testis is a network of channels at the back of the testis that collects sperm from the tubules. It is a passage, not a hormone source.
- **D** The tunica albuginea is the dense white capsule on the testis that divides it into lobules. It is connective tissue, not an endocrine tissue.

QUESTION 2 · COMPETENCY 141

Which statement describes the dartos muscle?

A. Smooth muscle under the scrotal skin that wrinkles the skin to cut heat loss

B. Skeletal muscle in the spermatic cord that pulls the testis up toward the body

C. A serous membrane that covers the front and sides of the testis

D. A dense white capsule that divides the testis into lobules

Answer: A. The dartos is a thin layer of smooth muscle in the subcutaneous tissue of the scrotum. When it contracts, the scrotal skin wrinkles and tightens, which reduces heat loss.

Why the others are wrong

- **B** That is the cremaster muscle, which is skeletal muscle from the internal oblique. It runs in the spermatic cord, loops around the testis, and raises it toward the warmer body wall.
- **C** That is the tunica vaginalis, the outer tunic, a serous membrane that came from the peritoneum.
- **D** That is the tunica albuginea, the inner tunic, whose septa divide the testis into lobules.

QUESTION 3 · COMPETENCY 142

Sperm leaving the tail of the epididymis enter which structure next?

A. Ejaculatory duct

B. Rete testis

C. Efferent ductules

D. Ductus deferens

Answer: D. The ductus deferens begins at the tail of the epididymis. It carries the sperm up the spermatic cord, through the inguinal canal, over the ureter, and behind the bladder to its ampulla.

Why the others are wrong

- **A** The ejaculatory duct comes later. It forms behind the bladder where the ampulla of the ductus deferens joins the duct of the seminal vesicle.
- **B** The rete testis is inside the testis, before the epididymis. Sperm pass through it on the way out of the testis.
- **C** The efferent ductules carry sperm from the rete testis into the head of the epididymis, so they come before the epididymis, not after it.

QUESTION 4 · COMPETENCY 142

The ejaculatory duct forms where the ductus deferens joins the duct of which gland?

A. Seminal vesicle

B. Prostate

C. Bulbourethral gland

D. Epididymis

Answer: A. Behind the bladder, the ampulla of each ductus deferens joins the duct of a seminal vesicle. Together they form the short ejaculatory duct, which passes through the prostate and empties into the prostatic urethra.

Why the others are wrong

- **B** The prostate does not form the ejaculatory duct. The ejaculatory duct passes through the prostate, and the prostate's own small ducts empty separately into the prostatic urethra.
- **C** The bulbourethral glands sit below the prostate and empty into the spongy urethra, well past the ejaculatory duct.
- **D** The epididymis is not a gland. It is the coiled duct on the back of the testis where sperm are stored and mature, and it leads into the ductus deferens.

QUESTION 5 · COMPETENCY 143

Which part of a sperm cell is packed with mitochondria?

A. Head

B. Middle piece

C. Principal piece

D. Acrosome

Answer: B. The middle piece is the first long stretch of the tail. Mitochondria wind in a spiral around the core of the tail there and make the ATP that powers the tail's movement. It is the only part packed with mitochondria.

Why the others are wrong

- **A** The head is almost all nucleus, with 23 chromosomes and almost no cytoplasm. It has no mitochondria to speak of.
- **C** The principal piece is the longest part of the tail and does most of the whipping, but the mitochondria that power it sit in the middle piece in front of it.
- **D** The acrosome is the cap over the front of the head. It holds enzymes that help the sperm get through the layers around the oocyte.

QUESTION 6 · COMPETENCY 144

Which erectile column surrounds the spongy urethra and forms the glans penis?

A. The two corpora cavernosa

B. The tunica albuginea

C. The prepuce

D. The corpus spongiosum

Answer: D. The corpus spongiosum is the single ventral column. It wraps around the spongy urethra along the body, its back end forms the bulb of the penis, and its expanded front end forms the glans.

Why the others are wrong

- **A** The corpora cavernosa are the two dorsal columns that lie side by side above the urethra. Their back ends form the crura of the root, not the glans.
- **B** The tunica albuginea is the dense capsule of the testis in this chapter. It is not an erectile column.
- **C** The prepuce is the foreskin, the loose fold of skin over the glans. It is skin, not erectile tissue.

QUESTION 7 · COMPETENCY 144

Which accessory gland adds the largest share of the volume of semen?

A. Prostate

B. Bulbourethral glands

C. Seminal vesicles

D. Epididymis

Answer: C. The paired seminal vesicles on the back of the bladder add an alkaline, fructose-rich fluid that makes up about 60 percent of the volume of semen.

Why the others are wrong

- **A** The prostate adds a milky fluid into the prostatic urethra that helps sperm stay active, but not the largest share.
- **B** The bulbourethral glands are pea-sized and add only a small amount of clear mucus into the spongy urethra, to neutralize urine traces and lubricate.
- **D** The epididymis is not an accessory gland. It is the duct on the back of the testis that stores and matures sperm.

QUESTION 8 · COMPETENCY 145

A patient with benign prostatic hyperplasia has a weak urine stream. Which structure does the enlarged prostate squeeze?

A. The bladder

B. The prostatic urethra

C. The ductus deferens

D. The membranous urethra

Answer: B. The prostate wraps around the prostatic urethra. When the gland enlarges, most often in the tissue right around the urethra, it squeezes that part of the urethra, so the stream weakens and the bladder may not empty fully.

Why the others are wrong

- **A** The prostate sits below the bladder and does not wrap around it. The bladder fails to empty because its outlet is squeezed, not because the bladder itself is compressed.
- **C** The ductus deferens runs behind the bladder and ends in the ejaculatory duct. It carries sperm, not urine, so squeezing it would not change the urine stream.
- **D** The membranous urethra passes through the pelvic floor muscles below the prostate. The prostate surrounds only the prostatic part of the urethra.